

AMAZING INTERNATIONAL AIRLINES INC.

ABSTRACT

To improve loyalty, Amazing International Airlines (AIAI) initiated a customer segmentation project. Our Exploratory Data Analysis (EDA) of three years of loyalty data **identified key behavioral patterns, seasonal trends, and significant data quality issues**. This analysis provides the clean dataset and foundational insights needed for future clustering.

INTRODUCTION

In the competitive airline market, loyalty is crucial. Amazing International Airlines (AIAI) has rich data but **lacks clear customer segmentation**. This project analyzes three years of loyalty and flight data to identify key behaviors and resolve data quality issues. Our **goal** is to establish a clean foundation for a **data-driven segmentation strategy and personalized marketing**.

DATA

The analysis utilized two primary datasets:

1. A **customer** file with 16,757 demographic and loyalty profiles;
2. A **flight activity** file spanning 36 months (2019-2021).

Initial exploration revealed significant data quality issues, such as duplicates and missing flight counts. **Key variables** like Customer Lifetime Value were also found to be highly skewed, indicating a small subset of customers drives most of the business.

METHODOLOGY

To guide our project, we adopted the **CRISP-DM** (Cross-Industry Standard Process for Data Mining) framework.

This structured approach ensured our analysis remained focused on clear **business goals**.

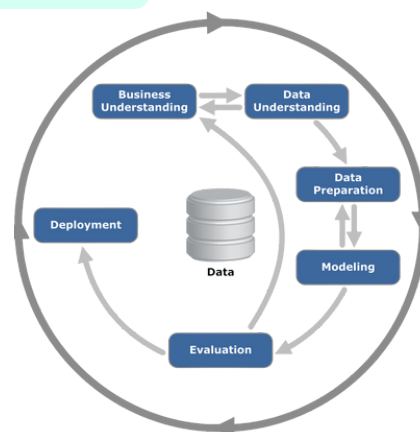


Fig.1 Diagram of CRISP-DM

METHODOLOGY

Our process began with **Business Understanding** to define our objective: analyzing airline data for key behavioral insights.

This was followed by **Data Understanding**, where we explored the dataset's key variables and patterns.

In the **Data Preparation phase**, we cleaned the data by handling missing values and inconsistencies to ensure reliable visualizations.

The core **Analysis & Visualization phase** then used exploratory techniques to identify significant patterns directly from the data.

RESULTS & DISCUSSION

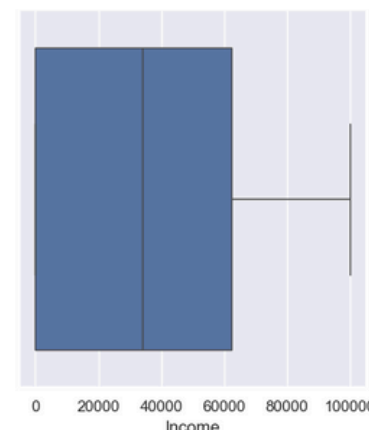


Fig.2 Box-Plot of Income var.

- Distribution highly (right) skewed;



Fig.3 Box-Plot of CLV var.

- Need for scaling and transformation in the Data Preparation stage

The distribution of Customer Income appears relatively uniform, with the median income centered around \$40,000-\$50,000 and the majority of customers falling within a predictable range.

In stark contrast, Customer Lifetime Value (CLV) is **highly skewed**. The vast majority of customers have a **low CLV**, clustered near the bottom of the scale.

Regarding our Flight's dataset, we were able to spot very clearly a distribution of the flights per month, as visualized on the Fig. 4, that can translate into useful insights.

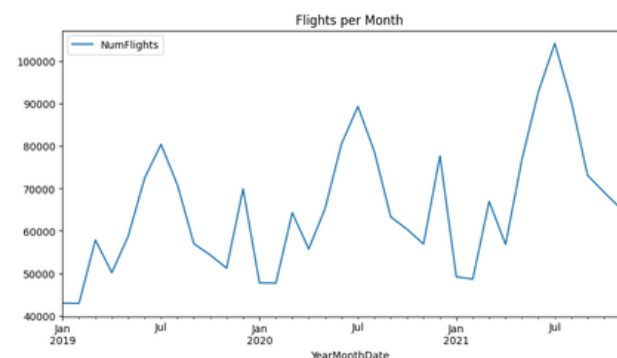


Fig.4 Plot of flights variance per month

RESULTS & DISCUSSION

Throughout the summer season, we can clearly delineate a trend where the AIA customers tend to travel more, and therefore bring higher revenues to the company. Special marketing campaigns for this specific time frame, could increase revenue and customer retention.

KEY INSIGHTS

CUSTOMER ACTIVITY IS HIGHLY POLARIZED:

- 1 Most loyalty members show long periods of inactivity (median monthly flights = 0), while a small group of high-frequency travelers drives the majority of flight volume.

STRONG SEASONALITY AND GROWTH TRENDS:

- 2 Flight activity **peaks in summer** and **increases year-over-year** (from 2019 to 2021) reflecting recovering demand and indicates the possibility of segmentation by **temporal dimension**.

LOYALTY ENGAGEMENT IS PRIMARILY ACCUMULATION-FOCUSED:

- 3 Customers rarely redeem their points and point balances typically increase over time. This indicates latent loyalty value, which may be an opportunity to optimize reward redemption strategies.

SOME HIGH-VALUE CUSTOMER ASYMMETRY:

- 4 Income and Customer Lifetime Value features are heavily right-skewed, with a small subset of customers holding disproportionately high economic value. This makes them critical drivers for differentiating premium customer segments.

CONCLUSIONS

The exploratory phase successfully fulfilled the **Business and Data Understanding** stages of the **CRISP-DM** process. Through **detailed EDA** were identified key **behavioral patterns, seasonality and data integrity issues** within AIAI's loyalty datasets. Future work will focus on advanced data preparation and clustering to uncover actionable customer segments that drive personalized retention strategies.

REFERENCES

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